Xiang Fang

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Education

University of Southern California Los Angeles，USA
Master, Applied Data Science **2024.09-2026.05**
Rutgers University New Jersey，USA
Bachlor，Business Analysis and Information Technology **2020.09-2024.05**

Skill

Skill：SQL、R、Excel 、Tableau、PowerBI、MySQL、Python、AWS、Hadoop、MongoDB、Spark、Google Colab、Firebase、DB Browser、Google API、Docker、Git、Figma、Vibe Coding、Codex
Language：English(Fluent)，Chinese(Native)

Job Experience

Data Analysis, *UnionPay Internationa,* Shanghai **2025.05– 2025.08**

- Participated in monitoring and analyzing abnormal transaction behaviors. Independently wrote complex SQL multi-table association and verification logic, achieved efficient batch extraction of abnormal transaction records, and shortened risk screening time by 20%.
- Optimized risk control rules based on the analysis results. Used SQL to calculate the interception success rate and false-positive rate of new rules in massive historical data. Participated in writing 12 advanced interception rules and increased the risk discovery rate by 10%.

Data Analysis, USC Annenberg Web Team, Los Angeles **2024.09-2025.01**

- In response to the low quality of raw campus safety data covering multiple campuses across the United States and containing heterogeneous sensors, led the use of Python (Pandas) and SQL for data cleaning and successfully improved the query response speed of the data dashboard by 30%.
- Participated in using SQLite to build a structured business database and used Docker to containerize the data service to ensure that the system was stable and available. Independently connected with Google Maps API to implement an interactive heatmap and achieved real-time data visualization support from the spatial dimension.

AI Agent, Koal Company，Shanghai **2024.05-2024.08**

- Independently designed and built a BI dynamic dashboard, integrated all sales data, and visually presented key performance indicators and seasonal trends. Actively communicated with the sales and operations teams across departments, unified business data standards, promoted the dashboard to be frequently used across departments in daily work, drove product strategy adjustments, and shortened the cross-team decision response cycle by 40%.
- Led the deployment of an internal HR intelligent question-and-answer assistant based on FASTGPT and Docker, transformed more than 500 company policy documents into a searchable knowledge base, and reduced the number of manual consultation tickets in the HR department by 60%.

Program Experience

Query Language ChatDB, Department of Computer Science, University of Southern California **2025.01-2025.05**

- **[Project Background]** In response to the problem that non-technical personnel find it difficult to directly extract multi-source heterogeneous tourism data, led the design of an "intelligent interactive data retrieval system," aiming to reduce the difficulty of cross-database data queries and shorten the data extraction cycle.
- **[Data Governance]** Collected and integrated more than 12,000 messy and heterogeneous data records from multiple mainstream tourism websites. Used Python (Pandas) / Spark to conduct distributed standardized cleaning and field alignment, successfully built a standardized MongoDB database, and improved subsequent data query and preparation efficiency by 50%.
- **[Intelligent Retrieval]** Used Docker to containerize and deploy database services and debugged the Ollama local large language model, achieved automatic conversion from natural language to standard SQL/NoSQL query statements and efficient extraction, increased the success rate of complex retrieval by 10%, and shortened the average query response time by 30%.
- **[Final Result]** The project was successfully implemented as a complete low-threshold data extraction tool, completely replacing the traditional inefficient operating model of "manually submitting requirements—technical personnel writing SQL and running data," and greatly improving the self-service data acquisition efficiency of non-technical business personnel.

TrailMind, Department of Computer Science, University of Southern California **2025.09-2025.12**

- **[Project Background]** In response to the problem that multi-source unstructured information in outdoor travel, such as weather, road conditions, and text guides, is fragmented and difficult to retrieve efficiently, led the design of an intelligent recommendation and navigation system, aiming to accelerate users' travel decision-making process through a fully automated pipeline.
- **[Data Governance]** Used Embedding technology to conduct unified vectorized cleaning of multi-source heterogeneous data such as CSV/PDF/HTML, and built an automated data pipeline based on LangChain to transform real-time environmental data into structured announcements, achieving a semantic retrieval accuracy rate of more than 90% and improving information distribution efficiency by 80%.

- **[Recommendation Strategy]** Designed a dialogue-driven recommendation engine and used the Redis high-speed caching mechanism to finely manage multi-turn conversation states and user intent context, controlled system latency at the millisecond level, and successfully shortened users' travel screening and decision-making time by 50%.
- **[Final Result]** Successfully implemented a fully automated intelligent navigation closed loop, achieved end-to-end efficiency improvement from "real-time cleaning of heterogeneous data" to "accurate recommendation through multi-turn conversations," and completely replaced the traditional manual information retrieval and organization model.